Homework 7

Due: Friday, October 30 at 11:59 p.m. PT

MATH 112, Fall 2026

Instructions

- As part of this homework, you must also submit the Mid-Semester Learning Reflection on Gradescope and complete the Google Course Feedback Form that is linked here
- Write your full name, "Homework 7," and the date at the top of the first page.
- Show all work and explain your reasoning. Write in complete sentences.
- Typeset your solutions in LaTeX if you can.
- Submit a single PDF file to Gradescope under the assignment "Homework 7." Submit the Mid-Semester Learning Reflection separately on Gradescope and complete the anonymous Course Feedback Form using the provided link.
- Use Gradescope to electronically match problems to pages in your PDF.
- Questions? Email the instructor or come to office hours; see Moodle.
- You are strongly encouraged to work together. Write up your own solutions.

Be sure to show your work. Organize it so that another MATH 112 student who has not seen the problems can follow your reasoning from line to line. Solutions that contain only an answer, without justification or supporting work, will not receive full credit.

Assignment 1 (graded for completion)

Complete the self-reflection form and upload it on Gradescope (separate assignment than this homework). You can either scan the completed form you were given in class or type your answers using the tex version of the form uploaded on Moodle. The form is also due October 30th at 11:59 PT.

Assignment 2 (not graded but important to do)

Complete the anonymous course feedback form - link here. The form is also due October 30th at 11:59 PT.

Problem 1 (15 points)

Polar form and multiplication

(a) For each of the complex numbers

$$z_1 = \sqrt{3} + i, \qquad z_2 = -1 + i, \qquad z_3 = -\sqrt{3} - i,$$

find its modulus and an argument in $[0, 2\pi)$, write it in polar form, and sketch it in the complex plane.

(b) Write z_1 and z_2 in polar form and use the rule for multiplication in polar form to find $z_1 z_2$. Give the modulus and an argument of the product, and verify your answer by multiplying in Cartesian form.

(c) Explain geometrically what multiplication by i does to a nonzero complex number. What does multiplication by -1 do?

Problem 2 (10 points)

Powers of complex numbers

(a) Find the value of

$$(1+i)^{20} - (1-i)^{20}.$$

(b) How many distinct values can

$$i^n + i^{-n}$$

take as n ranges over the positive integers? Justify your answer.

Problem 3 (15 points)

De Moivre's formula and roots of unity

(a) State de Moivre's formula. Use it to calculate

$$\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)^8$$

and write the answer in the form $a + bi$.

(b) Find all the third roots of unity, giving each in the form $a + bi$. Sketch them in the complex plane and explain why they form the vertices of an equilateral triangle.

(c) Explain why, in general, the nth roots of unity are n equally spaced points on the unit circle.

Problem 4 (20 points)

Modulus and geometry

(a) Let $z_1, \ldots, z_n \in \mathbb{C}$, where $n \geq 1$. Prove by induction that

$$|z_1 + \cdots + z_n| \leq |z_1| + \cdots + |z_n|.$$

Hint: Use the triangle inequality for two complex numbers.

(b) For $z_1, z_2 \in \mathbb{C}$, prove the *parallelogram identity*

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2.$$

(c) Among all complex numbers z satisfying

$$|z - 4i| = 2,$$

which has the smallest modulus, and which has the largest? Find both complex numbers and their moduli, and justify your answers.

Problem 5 (10 points)

Distances on the complex plane

Define $d : \mathbb{C} \times \mathbb{C} \to \mathbb{R}$ by

$$d(z, w) = |\operatorname{Re}(z - w)| + |\operatorname{Im}(z - w)|.$$

Prove that d is a distance on $\mathbb{C}$ by verifying positivity, symmetry, and the triangle inequality.

Problem 6 (30 points)

Conjugation and its consequences

(a) Let

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad a_0, \ldots, a_n \in \mathbb{R}.$$

Prove, in the following steps, that nonreal roots of p occur in conjugate pairs.

(i) Show that $\overline{z^k} = (\overline{z})^k$ for every integer $k \geq 0$. Explain why $\overline{a} = a$ for every $a \in \mathbb{R}$.

(ii) Show that $\overline{p(z)} = p(\overline{z})$ for every $z \in \mathbb{C}$. Indicate where the assumption that the coefficients are real is used.

(iii) Deduce that $p(z) = 0$ implies $p(\overline{z}) = 0$. Why are z and $\overline{z}$ distinct when z is nonreal?

(iv) For $z \in \mathbb{C} \setminus \mathbb{R}$, expand $(x - z)(x - \overline{z})$ and express its real coefficients in terms of $\operatorname{Re}(z)$ and $|z|$.

(v) Find a quadratic polynomial with real coefficients whose roots are $2 + 3i$ and $2 - 3i$. Explain why a real quadratic polynomial cannot have exactly one nonreal root.

(b) Let $z \in \mathbb{C}$ with $|z| = 1$. Prove that

$$\overline{z} = \frac{1}{z}.$$

(c) If $|z_1| = \cdots = |z_k| = 1$, prove that

$$|z_1 + \cdots + z_k| = \left| \frac{1}{z_1} + \cdots + \frac{1}{z_k} \right|.$$